FUTURE SELF-PROOF ELICITATION MECHANISMS

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Abstract. We introduce and study, both theoretically and empirically, the notion of 'future self-proof' mechanisms, those in which boundedly rational individuals do not need to rationally forecast their own future actions (or those of others). We focus on elicitation mechanisms due to their importance and relative simplicity. We first construct an obviously strategy-proof (OSP) elicitation mechanism, showing that it greatly outperforms the standard Becker-Degroot-Marschak (BDM) mechanism, but also sometimes fails due to the fact that it is not future self-proof. Motivated by recent experimental findings, we then define a new notion of future self-proof mechanisms, which we call game-structure obvious (GSO). We find empirically that a GSO mechanism, which resembles a multiple price list, outperforms the OSP mechanism, but has its own source of mistakes, suggesting an inherent tradeoff in designing robust mechanisms.

1. Introduction

Mechanism design is perhaps the most practical tool to come out of microeconomic theory, with applications ranging from auctions to assignment problems (e.g. assigning students to schools). However, mechanisms only work to the extent that people understand them, and behavioral economics has produced overwhelming evidence that people are severely cognitively constrained. Even dominant strategy mechanisms do not work as well as theory would suggest (e.g. overbidding in second price auctions (Kagel et al. (1987) and Kagel and Levin (1993) and untruthful reporting in strategy-proof matching mechanisms (Chen and Sönmez (2006)). In fact, even individual elicitation mechanisms in which people do not have to worry about the behavior of others often do not function well (e.g. untruthful reporting of valuations (Cason and Plott (2014) and of beliefs (Danz, Vesterlund, and Wilson (2020)). Given these limitations, a nascent literature has turned to developing

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robust mechanisms that ensure truthful revelation from even boundedly rational individuals (e.g. Li (2017a) and Pycia and Troyan (2021)).

In this paper, we study mechanism design for boundedly rational players both theoretically and empirically, focusing on the simplest mechanisms - elicitation mechanisms. Elicitation mechanisms are crucial tools for economists, allowing us to elicit valuations, certainty equivalents (to study risk preferences), and beliefs (to study belief updating in decision-making and strategic games), and to study classic anomalies such as the endowment effect. They are also relatively simple because, as individual decision-making problems, they allow us to rule out failures of strategic reasoning and other-regarding preferences as sources of deviations from theoretical predictions when studying them empirically.

A natural starting point for an elicitation mechanism that is robust to bounded rationality is Li's (2017) concept of obviously strategy-proofness (OSP). OSP is a desirable property in that players do not need to perform some types of contingent reasoning. In a second-price sealed bid auction (which is strategy-proof but not OSP), for each potential bid, players have to think through all of the hypothetical bids made by others in order to identify the scenarios in which they will win and lose, and then choose the optimal bid accordingly. Instead, in an ascending clock auction that is OSP, if the current price is below a player's valuation, then the best possible outcome from quitting now is no better than the worst possible outcome from staying in the auction and later quitting at the player's valuation. In this case, each player does not have to compare outcomes across sets of hypothetical bids of the other players, greatly reducing the amount of reasoning required.

Importantly, however, OSP mechanisms still require players to correctly forecast their *own* future actions at all possible contingencies and then to choose the best current action given this plan.[1] In an ascending clock auction, for example, continuing at prices below a player's valuation can be sub-optimal if the player believes that she would also stay in for prices above her valuation. Given that people might fail to perform the contingent reasoning required to correctly forecast their own future actions (and we provide evidence that they do), we are motivated to ask whether or not mechanisms that do not require such reasoning exist.

We call such mechanisms future self-proof (FSP) mechanisms - mechanisms in which players do not need to worry about the actions of their 'future selves'. FSP mechanisms can be constructed in two ways. The first way comes from Pycia and Troyan's (2021) generalization of OSP mechanisms - strongly obviously strategy proof mechanisms (SOSP). In an SOSP mechanism, the worst case payoff from the

[1]The OSP mechanism we introduce actually satisfies the stronger one-step forward-looking (OSF) property of Pycia and Troyan (2021): players only need to look forward one period rather than many, but this still requires contingent reasoning.

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dominant action must be better than the worst case payoff from any deviation, given any action one's future self (or others) might take.

Our second, novel way of constructing an FSP mechanism, focuses directly on the structure of the game that makes up the mechanism, rather than the sets of payoffs. We call such mechanisms *game-structure obvious* (GSO), motivating our definitions using recent experimental evidence that studies contingent reasoning failures (Martínez-Marquina, Niederle, and Vespa (2019) and Esponda and Vespa (2021)). GSO mechanisms are simple for the same reason that playing second in a two-player, two-action game is easier than playing simultaneously: rather than thinking through the best response for each hypothetical action of the other player, the second player only has to best respond to the *actual* action taken by the first. More generally, a mechanism is GSO if a player can easily identify the best action at each decision node without forming correct beliefs about her other actions and the actions of others. In the context of elicitation mechanisms, GSO mechanisms take the form of a dynamic version of a multiple price list (MPL).

In our first theoretical result, we show that it is impossible to design a binary allocation mechanism that is SOSP (and elicits more than two types). Thus, the only possible FSP elicitation mechanism is a GSO mechanism. In our second result, we show that GSO elicitation mechanisms cannot simultaneously be OSP, raising the question of which does better in practice.[2] If people rationally condition on their own future actions, then OSP mechanisms should perform better because the dominant action is obvious in the sense of best/worst case payoffs. If they don't, GSO mechanisms may perform better because of the fact that they are future self-proof.

To answer this empirical question, we design an experiment with two main goals: (i) to compare the performance of a standard elicitation mechanism to that of its OSP and GSO counterparts, and (ii) to determine whether or not failures in the OSP mechanism can be attributed to it not being FSP. To achieve the first goal, we elicit subjects' willingness-to-accept (WTA) for a fixed item of $1.00 (following Cason and Plott (2014)) using the standard BDM mechanism (Becker, DeGroot, and Marschak (1964) and two novel mechanisms that we construct (OSP and GSO). In order to obtain the cleanest possible comparison, in each of our three treatments (BDM, OSP, and GSO), a subject completes exactly one elicitation task with one of the mechanisms, in a between-subjects design. To achieve the second goal, we designed a diagnostic treatment, and also asked subjects a single incentivized question after

[2]GSO mechanisms are also distinct from the partial obviously strategy-proof mechanisms of Zhang and Levin (2021). In particular, our GSO elicitation mechanism is not partial obviously strategy-proof.

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the elicitation task to see whether or not they understood the payoff consequences of their consequential choice in the OSP mechanism.

The two dynamic mechanisms, OSP and GSO, although having different theoretical properties, look very similar. Each mechanism runs with a clock price that starts at some high value ($1.50 in our implementation) and decreases by $0.05 in each subsequent period. The subject chooses between the price and the $1 item each period. At some randomly selected period, the mechanism terminates and the subject receives their choice in that period. In the GSO mechanism, a subject faces the same choices in the following rounds irrespective of her choices in previous rounds. The key difference in the OSP mechanism is that, in each round, choosing the item ends the game so that future choices are only made when the price is chosen. In both mechanisms, the weakly dominant strategy is to choose the item after the price reaches the subject's value of the item, revealing her valuation.

Our first main result is that the OSP mechanism greatly outperforms traditional BDM: the fraction of rational decisions (those that reveal a valuation of $1) more than doubles from 15% to at least 35%.[3] This result provides evidence, in addition to Li (2017a), that OSP mechanisms do indeed outperform their non-OSP equivalents. However, in the OSP mechanism, we observe that more than 70% of the elicited valuations that are inconsistent with rationality exceed $1, implying that most subjects exited before the price reached the rational valuation. One possible explanation for this finding is that the OSP mechanism is not FSP. Subjects may have *future self-concerns*: those who do not correctly forecast their future actions (quitting at the valuation) may worry about ending up with less than $1, and therefore try to pre-empt this possibility by choosing the item too soon.

To test this potential explanation, we conduct a diagnostic treatment using a future self-proof (SOSP) version of the OSP mechanism. This mechanism is identical to the OSP mechanism except that subjects are forced to choose the item when the clock price reaches the monetary value of item ($1). Given that subjects know this ex ante, it removes all uncertainty about their future actions.[4]

Comparing the fraction of irrationally high valuations (those greater than $1) in the SOSP treatment to that in the OSP treatment provides our second main result. We find strong evidence that subjects have difficulty correctly forecasting their future actions: the fraction of high valuations is cut in half (from 55% to 27% or 43 to 22% depending on the sample) in the SOSP treatment. Furthermore, using the incentivized question, we show that subjects that choose the item at a price greater than $1 understand that they are potentially sacrificing money by doing so, further

[3]Due to indifference at $1, we treat valuations of both $0.95 and $1.00 as rational.

[4]Because this mechanism requires the experimentalist to know subjects' valuations ex ante, it is useful for diagnostic purposes only.

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strengthening the claim that they are trying to pre-empt future potential losses. This finding provides the first available evidence (to our knowledge) that future self-concerns may pose a problem for mechanisms (including OSP mechanisms) more generally.

Given that future self-concerns are important in the OSP mechanism, our final question is whether or not a GSO mechanism, which is FSP, outperforms it. We find that the answer is yes: the fraction of rational decisions increases from 35 to 40% in OSP to 50% in GSO, an improvement of at least 25%.[5] Thus, at least in this setting, it seems that making the structure of the game obvious is more important than making the mechanism obvious in the OSP sense (worst case payoff from the dominant strategy is greater than the best case payoff from any deviation). However, we find that GSO mechanisms have their own, unique problem. By having subjects make very similar decisions at each point in time, a GSO mechanism intuitively makes the dominant strategy obvious. But, it also requires subjects to make similar decisions when faced with a sequence of similar problems. We find, consistent with the large literature on MPL elicitation schemes, that they don't: 35-40% of subjects switch multiple times between the item and price, rendering their valuations indeterminate. Thus, although our GSO mechanism outperforms the OSP mechanism, it suffers from a well known, but not yet well understood problem, the source of which we speculate about in the discussion.

Our theoretical contributions build on the recent literature interested in designing mechanisms that are robust to bounded rationality, specifically Li (2017a) and Pycia and Troyan (2021).[6] We apply results in this literature to elicitation schemes, and introduce the novel idea of GSO mechanisms. Our formal definition of GSO is motivated by experimental results that show that subjects perform worse in simultaneous games than equivalent sequential games (Carrillo and Palfrey (2011), Esponda and Vespa (2014), Magnani and Oprea (2017), Ngangoué and Weizsäcker (2021)) and, more particularly, experiments that study failures of contingent reasoning (Martínez-Marquina, Niederle, and Vespa (2019) and Esponda and Vespa (2021)).

Elicitation schemes have been studied and compared in several papers. Cason and Plott (2014) were the first to clearly demonstrate the problem with BDM mechanisms by eliciting valuations for an object of known value. Martin, Munoz, and

[5]In GSO, any subject that switches multiple times is counted as irrational.

[6]See also Zhang and Levin (2017), Li (2017b), Mackenzie (2020), and Zhang and Levin (2021). More distantly related is the literature studying mechanisms in matching applications. See Troyan and Morrill (2020) and the references therein.

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Rodriguez (2020) replicate their finding and show that reframing BDM by providing contingency tables improves its performance.[7] Bartling, Engl, and Weber (2015) also use BDM to elicit valuations for a known dollar amount, but frame the BDM mechanism as a price list. This implementation is similar to our OSP mechanism except that the randomly chosen price is only revealed after subjects make all choices. Unlike our experiment, they do not compare to a static implementation of BDM, instead focusing on the endowment effect.

Also related are the experiments by Li (2017a) and Breitmoser and Schweighofer-Kodritsch (2021) that compare second-price sealed bid auctions to clock auctions, similar to our comparison of BDM and OSP elicitation mechanisms, except that strategic ambiguity about other's play may play a role in interpreting their results. Both studies find that clock auctions perform better, consistent with our results that OSP outperforms BDM. Breitmoser and Schweighofer-Kodritsch (2021) also conduct several treatments to decompose the differences between the bid auction and the clock auction, showing that, after experience, the OSP property of the clock auction is not necessary for good performance. However, in initial behavior, which would seem to be most relevant for elicitation tasks that are typically only performed once, they find that the clock auction does better.[8]

Ivanov, Levin, and Niederle (2010) and Bosch-Rosa and Meissner (2020), respectively, show that subjects do not always best respond to their own actions when taken in the past or when taken simultaneously. We add to this evidence by showing that subjects also do not fully understand their future actions, which is particularly important in that doing so is necessary for OSP mechanisms to function well. Hao and Hauser (2012) elicit probabilities using the two mechanisms suggested by Karni (2009), one of which is static and one of which uses a clock. Although a similar comparison to our comparison of BDM and OSP, their clock mechanism is not OSP (Tsakas, 2019).

The rest of the paper is organized as follows. Section 2 introduces introduces formal definitions and presents the theoretical results. Section 3 describes the experimental design and Section 4 reports our empirical findings. Section 5 concludes with a discussion.

[7]Our OSP mechanism is similarly reframed, but providing contingency tables on their own does not make the BDM elicitation mechanism OSP.

[8]We would argue that one reason for making mechanisms 'obvious' is so that they work without extensive experience and feedback, an aspect of mechanisms that has been of interest in previous work (e.g. Chen and Plott (1996), Chen and Tang (1998), Chen and Gazzale (2004), and Healy (2006)).

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2. Theory

2.1. **Notation for Extensive form games.** Let every player's valuations for an item (private type) be drawn from a set $\mathcal{V} = \{v_1, v_2, ..v_n\}$ with $|\mathcal{V}| = n$. We say that type i has valuation $v_i \in \mathcal{V}$ where, without loss, $v_k > v_l$ for $k > l$.

An extensive form-game with consequences in X is a tuple $(N, H, \succ, \mathcal{A}, P, (\mathcal{I}_i, u_i)_{i \in N})$, where N is the set of players, H is a set of histories, P maps histories to the player who moves at that history, $(\mathcal{I}_i)_{i \in N}$ partitions the set of histories into information sets for the players, $\mathcal{A}$ maps information sets (and therefore non-terminal histories) to available actions, and u_i maps each terminal history to outcomes in X for player i. $(H, \succ)$ forms an arborescence for a partial order $\succ$ on the set of histories. Thus, we write $h_1 \succ h_2$ whenever $h_1 \in H$ succeeds $h_2 \in H$. Similarly, for $I_i, I_i' \in \mathcal{I}_i$, we say that $I_i \succ I_i'$ if there exists $h_1 \in I_i$ and $h_2 \in I_i'$ such that $h_1 \succ h_2$. When making a decision at any information set, I_i', we refer to decisions made by i at future information sets, $I_i \succ I_i'$, as being made by i's *future self.*

A strategy, S_i, for a player i is a function that maps each of i's information sets into actions available at that information set. Let $\mathcal{S}_{-i}$ be the set of all strategies that can be played by non-i players, which, for elicitation tasks, includes only Nature. We assume perfect recall[9], finite depth, and $|\mathcal{A}(I_i)| < \infty$ for all I_i.

Binary allocation mechanisms. A mechanism is an extensive game form. In our experiment, we focus on *binary allocation mechanisms*, where a single item is either allocated to a player or not. For example, consider the BDM mechanism for elicitation of a subject's WTA. Nature randomly draws a number, v_r, and a subject reports their valuation, v. BDM then implements the choice rule:

$$\underbrace{f(v)}_{\text{seller's outcome}} = \begin{cases} \underbrace{(1,0)}_{\text{(seller keeps item, gets no money)}} & \text{if } v_r \leq v \\ \underbrace{(0, v_r)}_{\text{(seller sells item, gets } v_r\text{)}} & \text{if } v_r > v \end{cases}$$

Strongly Obviously Strategy-proof Elicitation Mechanisms (SOSP). Following Li (2017), a strategy S_i is *obviously dominant (OD)* if the worst possible payoff to a player from playing S_i is at least as good as the best possible payoff from any deviation, where the best/worst case outcomes are determined over all possible future actions of other agents (S_{-i}) and all possible choices of Nature. As S_i is a full contingent plan for the whole game, to discover such a strategy, players must be able to

[9]On the path from the initial history to a history h, list in chronological order the information sets I_i that were encountered and the actions i chose in each. Call this list the experience, $X_i(h)$, of i in history h. A game has perfect recall if all histories in the same information set have the same experience.

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forecast their own future actions at all future information sets. Pycia and Troyan's (2021) *Strongly Obviously Dominant (SOD)* concept strengthens *OD* by requiring that the best/worst case outcomes be determined over all possible future actions of other players (S_{-i}), Nature, *and a player's own future actions.* Mechanisms are Obviously Strategy Proof (OSP) or Strongly Obviously Strategy Proof (SOSP), respectively, when an *obviously dominant* or *strongly obviously dominant* strategy exists for every type of every player, and together these strategies implement the choice rule (see Li (2017) and Pycia and Troyan (2021) for formal definitions of OSP and SOSP, respectively). Thus, SOSP mechanisms are FSP in the sense that they do not require players to think through their own future actions. However, the SOSP concept places strong restrictions on the set of social choice functions that can be implemented. In particular, we show that the SOSP property severely limits what can be identified via an elicitation mechanism.

For any mechanism to elicit all player types, it must be the case that each player-type plays an (observationally) different strategy. Formally,

Definition. We say that a SOSP mechanism identifies all relevant types for participant i if there exists a set of $|\mathcal{V}|$ strategies (messages) $\{S_j : j \in \mathcal{V}\}$ such that strategy S_k is *strongly obviously dominant* (SOD) for type v_j if and only if $k = j$.

Pycia and Troyan (2021) show that any SOSP mechanism takes the form of a sequential price game (Theorem 5). In eliciting the valuation from a single player i, this restriction implies that any SOSP mechanism offers a *single* payoff-relevant action to player i. For SOSP-identification, this single choice must identify all of i's possible types. Our first result is an impossibility result: for more than two types, $|\mathcal{V}| > 2$, there does not exist a *binary allocation* mechanism that is SOSP (with a single payoff-relevant action) and identifies all types. In fact, we prove a more general result. Consider a general mechanism that can assign $p \in \{0, 1, 2, ..m\}$ units of the item to any participant - the binary allocation mechanism being the special case that can only assign up to $m = 1$ units. We show that for an SOSP mechanism to identify types in $\mathcal{V}$, m has to be at least as large as $|\mathcal{V}| - 1$.

Proposition 1. *i) When $m < |\mathcal{V}| - 1$, there exists no SOSP allocation mechanism that transacts at most m units of the item and identifies all relevant types.*
ii) When $m = |\mathcal{V}| - 1$, a SOSP allocation mechanism exists that identifies all relevant types using at most m units of the item.

Proposition 1 shows that, for practical purposes, SOSP elicitation mechanisms do not exist. If only one item is to be allocated, the mechanism designer can elicit up to two possible types. So, if the designer wants to elicit a player's valuation of an item, she can only determine whether it is above or below some amount,

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which is practically not very useful. If the designer wants to more accurately determine the type, she needs to allocate more items. In willingess-to-pay (WTP) or willingness-to-accept (WTA) mechanisms, the designer can allow the player to transact multiple units but this is likely to affect the player's WTA/WTP.[10] Given that SOSP elicitation mechanisms are so restrictive, we seek another way to define FSP mechanisms.

Game Structure Obviousness (GSO). SOSP mechanisms are FSP because the payoffs associated with the prescribed action at I_i' are 'obviously' better than those for any other action, irrespective of how future selves act at a future information set, $I_i \succ I_i'$. But, SOSP mechanisms still require a player to evaluate the maximum and minimum payoffs, and thus foresee the set of all of the possible payoffs following each action. If the structure of the mechanism is complex, the sets of payoffs associated with each action may themselves be difficult to identify. Instead, one can think of defining an 'obvious' mechanism as one in which a player can easily determine how to play now without determining her future actions or the actions of others. We use this intuition to formally define *game-structure obvious* mechanisms. We will also define a notion of *weak dominance against future selves*: the payoffs associated with the prescribed action at I_i' are better than those for any other action, irrespective of how future selves act at a future information set, $I_i \succ I_i'$, as long as they play strategies from a naturally restricted set.

We say that a mechanism is GSO for player i if it satisfies four properties that we motivate and define in the following. The first property is that, every non-i player must have moved before i's earliest move. Formally,

Definition. i is the *last mover* if $h_j \not\succ h_i$ whenever $P(h_j) \neq i$ and $P(h_i) = i$.

We have not assumed that the game has perfect information, so moving last is not necessarily informationally advantageous. But, by moving last, i does not have to worry about how her opponents might play differently in the future based on her playing different actions now. For games in which the only player other than i is Nature, being the last mover implies that Nature moves first (although, again, i need not observe Nature's move).

The second property guarantees identical continuation games for player i irrespective of i's current choice. We will call this property *similar future actions and outcomes*. Intuitively, we require such a property so that a player can treat her current action as being 'independent' of her future actions. To define it, we first

[10]For example, one can elicit WTA or WTP using a multiple price list in which every choice is paid. This mechanism would be SOSP because the number of items allocated is equal to the number of choices, say n. But, for WTA, it would require the player to have n items to begin with and for WTP, she might end up with n items.

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need to define *pre-terminal histories* as those histories within an information set that immediately lead to a terminal history.

Definition. At any information set I_i, the subset of *pre-terminal histories* is defined as

$$PT(I_i) = \{h \in I_i : \nexists h', h'' \text{ such that } h'' \succ h' \succ h\}$$

Let $a_1, a_2 \in A(h)$ be two actions available to i at some information set I_i^0. We will use $I_i^0|a_1$ and $I_i^0|a_2$ to denote the collection of i's information sets available at the continuation games following the actions a_1 and a_2, respectively. Such future information sets are available only when not all histories in the current information set are *pre-terminal* (i.e, $I_i \neq PT(I_i)$). Intuitively, we will say that i experiences similar future actions, if irrespective of her current actions, she faces similar information sets and actions in the continuation games. For any strategy S of player i , and $a_1, a_2 \in \mathcal{A}(I_i)$, let a_1S (and a_2S) be the modified strategy where she plays a_1 (and a_2) at information set I_i, and plays according to S at all other information sets. Let $z_i(h, S_i, S_{-i})$ be the outcome for player i obtained in the game when we start from history h and play proceeds according to (S_i, S_{-i}). Let $u_i(h, S_i, S_{-i}, v_i)$ be type $v_i \in \mathcal{V}$'s utility from the outcome obtained when we start at history h, and thereafter play proceeds according to (S_i, S_{-i}). We denote the initial history h_ϕ. We say that action a_k leads from one information set I^k to another information set I^{k+1}, if a_k leads from a history in I_k to a subsequent history in I_{k+1}.

Definition. We say that a game in which i is the last mover satisfies *similar future actions and outcomes for* i if at any information set $I_i \in \mathcal{I}_i$ with $I_i \neq PT(I_i)$, and for any two actions $a_1, a_2 \in \mathcal{A}(I_i)$, there exists a bijective mapping $f_{a_1,a_2} : I_i^0|a_1 \to I_i^0|a_2$, and, corresponding bijective mappings $g_{a_1,a_2,I^k} : \mathcal{A}(I^k) \to \mathcal{A}(f_{a_1,a_2}(I^k))$ for all $I^k \in I_i^0|a_1$, such that:
i) (similar future actions) for $I^k, I^{k+1} \in I_i^0|a_1$, action a_k leads from I^k to I^{k+1} if and only if action $g_{a_1,a_2,I^k}(a_k)$ leads from $f_{a_1,a_2}(I^k)$ to $f_{a_1,a_2}(I^{k+1})$.
(ii) (similar future outcomes) fixing any two strategies, S^{r1} and S^{r2}, that play similar actions at similar future information sets, i,e,

$$\underbrace{S^{rj}\left(f_{a_1,a_2}(I^k))\right)}_{\text{action at info set which is similar to } I^k} = \underbrace{g_{a_1,a_2,I^k}}_{\text{similarity mapping for actions}} \left(\underbrace{S^{rj}(I^k)}_{\text{action at } I^k}\right) \forall I^k \in I_i^0|a_1$$

for $j = 1, 2$, at all histories $h \in I_i$, one of the following two conditions hold. Either

$$\underbrace{z_i(h, a_1S^{r1}, S_{-i}^1) = z_i(h, a_1S^{r2}, S_{-i}^2) \quad \forall a_1 \in \mathcal{A}(I_i),\ \forall S_{-i}^1, S_{-i}^2 \in \mathcal{S}_{-i}}_{\text{future actions are irrelevant to the outcome}}$$

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or for $j = 1, 2$,

$$\underbrace{z_i(h, a_1 S^{rj}, S^1_{-i}) = z_i(h, a_2 S^{rj}, S^2_{-i}) \quad \forall a_1, a_2 \in \mathcal{A}(I_i),\ \forall S^1_{-i}, S^2_{-i} \in \mathcal{S}_{-i}}_{\text{outcome under any future action is unaffected by current actions}}$$

(i) defines similar future actions: even after disparate current actions, there exists a way in which all of the information sets reached thereafter, and the actions available therein, are 'similar'. f_{a_1,a_2} maps similar information sets to each other and g_{a_1,a_2,I^k} maps similar future actions across those similar information sets. (ii) says that, depending on the history, either future actions are irrelevant to the outcome, or that the outcome after a future action is unaffected by current actions, as long as some restricted strategy (S^{r1} or S^{r2}) is played in the future. Simply put, a player need not worry that her current action is going to change the game she faces in the future. In such games, players can ignore the contingencies that arise after their current actions, and instead make narrowly-framed comparisons, making accounting for future contingencies significantly easier. Given *similar future actions and outcomes* only imposes restrictions for information sets that allow for future continuation ($I_i \neq PT(I_i)$), games in which i moves last at a single information set trivially satisfy the property.

We use the similarity mapping used in defining *similar future actions and outcomes* to define a *restricted* strategy for i, which uses similar strategies at similar future information sets.

Definition. In a game in which i is the last mover and which satisfies *similar future actions and outcomes* for i, a strategy S_i for player i is *restricted* from $I_i \in \mathcal{I}_i$ onward if

$$\underbrace{S_i\left(f_{S_i(I_i),a_2}(I^k)\right)}_{\text{action at info set which is similar to } I^k} = \underbrace{g_{S_i(I_i),a_2,I^k}}_{\text{similarity mapping for actions}} \left(\underbrace{S_i(I^k)}_{\text{action at } I^k}\right)$$

for any $a_2 \in \mathcal{A}(I_i)\backslash\{S_i(I_i)\}$ and any future information set $I^k \in I_i|S_i(I_i)$ where i moves.

Let $\mathcal{S}^r(I_i)$ be the set of all i's strategies that are *restricted* from $I_i \in \mathcal{I}_i$ onward. Using *pre-terminal histories* and *restricted strategies*, we can introduce the concept of *dominance against future selves*. A dominant strategy against future selves exists when the best response at any information set is independent of what one's future selves and others play, as long as one's future selves play similarly at similar information sets. Thus, a player does not need to determine which strategy is played at future information sets: assuming any $S^r_i \in \mathcal{S}^r(I_i)$ suffices.

Definition. Fix a game in which i is the last mover and which satisfies *similar future actions and outcomes*. A strategy S_i in this game is *weakly dominant against future*

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selves (WDFS) for type v_i if $\forall I_i$,
i) when $PT(I_i) = I_i$, for $S_i(I_i) = a$,

$$u_i(h_\phi, a, S_{-i}, v_i) \quad \geq \quad u_i(h_\phi, a', S_{-i}, v_i) \;\; \forall h \in I_i, \forall a' \in \mathcal{A}(I_i), \forall S_{-i}$$

ii) when $PT(I_i) \neq I_i$, for $S_i(I_i) = a$,

$$\begin{aligned} u_i(h_\phi, aS_i^r, S_{-i}, v_i) \quad &\geq \quad u_i(h_\phi, a'S_i^r, S_{-i}, v_i) \\ &\qquad \forall h \in I_i, \forall a' \in \mathcal{A}(I_i), \forall S_{-i}, \quad \underbrace{\forall S_i^r \in \mathcal{S}^r(I_i)}_{\text{any restricted strategy}} \end{aligned}$$

We now show that a WDFS strategy always exists for games that satisfy *similar future actions and outcomes,* and a *no cycles* condition that we define below. Furthermore, such a WDFS strategy can be discovered by a simple algorithm: choose actions that are undominated at the *payoff-relevant* histories, assuming some *restricted* strategy is played in the future. A history is payoff-irrelevant for type v_i if all available actions lead to identical payoffs. The set of *payoff-relevant* histories is the complement of the set of payoff-irrelevant histories:

Definition. At any information set I_i in a game in which i is the last mover, we denote the subset of all histories where actions are *payoff-relevant* for type v_i as $\mathbb{I}(I_i) \subseteq I_i$, such that

$$\begin{aligned} u_i(h, aS_i^r, S_{-i}, v_i) \quad &= \quad u_i(h, a'S_i^r, S_{-i}, v_i) \\ &\qquad \forall h \in I_i \backslash \mathbb{I}(I_i), \forall a, a', \in \mathcal{A}(I_i), \forall S_{-i}, \forall S_i^r \in \mathcal{S}^r(I_i) \end{aligned}$$

Definition. A game in which i is the last mover and which satisfies *similar future actions and outcomes*, additionally satisfies the *no cycles* condition for i if
i) When $PT(I_i) = I_i$,

$$\begin{aligned} u_i(h, a_1, S_{-i}^1, v_i) > u_i(h, a_2, S_{-i}^1, v_i) \iff u_i(h', a_1, S_{-i}^2, v_i) \geq u_i(h', a_2, S_{-i}^2, v_i) \\ \forall h, h' \in I_i, \forall a_1, a_2 \in \mathcal{A}(I_i), \forall S_{-i}^1, S_{-i}^2 \in \mathcal{S}_{-i}, \forall v_i \in \mathcal{V} \end{aligned}$$

ii) when $PT(I_i) \neq I_i$,

$$\begin{aligned} \underbrace{u_i(h, a_1 S^{r1}, S_{-i}^1, v_i) > u_i(h, a_2 S^{r1}, S_{-i}^1, v_i)}_{a_1 \text{vs } a_2 \text{against } S_{-i}^1 \text{ with } S^{r1} \text{played in the future}} \iff \underbrace{u_i(h', a_1 S^{r2}, S_{-i}^2, v_i) \geq u_i(h', a_2 S^{r2}, S_{-i}^2, v_i)}_{a_1 \text{vs } a_2 \text{against } S_{-i}^2 \text{ with } S^{r2} \text{played in the future}} \\ \forall h, h' \in I_i, \forall a_1, a_2 \in \mathcal{A}(I_i), \forall S^{r1}, \ldots \\ S^{r2} \in \mathcal{S}^r(I_i), \forall S_{-i}^1, S_{-i}^2 \in \mathcal{S}_{-i}, \forall v_i \in \mathcal{V} \end{aligned}$$

The *no cycles* condition says that if a_1 is a strictly better than a_2 for some situation (history), then there cannot be other situations where a_2 is strictly better than a_1, thus ruling out a cycle. In the presence of non pre-terminal histories, we require that some restricted strategy (S^{r1} or S^{r2}) is played in the future. The independence over opponents' strategies implies that a_1 and a_2 can be ranked consistently

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for all actions taken by non-i players. The independence over histories additionally implies what i has played in the past does not matter for the ranking.

Proposition 2. *If i is the last mover, and the game satisfies similar future actions and outcomes and the no cycles condition, then a weakly dominant against future selves (WDFS) strategy S_i^* exists for every type v_i. Furthermore, S_i^* can be constructed by the following algorithm: at every information set I_i, fix some restricted future strategy $S^{r1} \in \mathcal{S}^r(I_i)$ and then select an action that best responds against all opponents' strategies at all payoff-relevant histories: $S_i^*(I_i) \in T(I_i)$ where,*

$$T(I_i) = \{a \in \mathcal{A}(I_i) : u_i(h, aS^{r1}, S^1_{-i}, v_i) \geq u_i(h, a'S^{r1}, S^1_{-i}, v_i) \;\; \forall a' \in \mathcal{A}(I_i), \; \underbrace{\forall h \in \mathbb{I}(I_i)}_{payoff\ relevant}, \forall S^1_{-i}\}$$

We could define a weak notion of GSO using *similar future actions and outcomes* and the *no cycles* conditions given that these, together with i being the *last mover*, would imply the existence of a WDFS strategy for player i. A WDFS strategy is obvious in the sense that a player can identify it by simply focusing on the *payoff-relevant* histories each time she is called to move. However, in complex games, the set of *payoff-relevant* histories may be large, so that the need to only focus on them may not obviate the need for contingent reasoning. In fact, experimental evidence suggests that this is the case. We present this evidence in motivating our third property, which strengthens what GSO requires.

First, BDM trivially satisfies *similar future actions and outcomes* with i as the *last mover* because i moves only once. Thus, in BDM, the WDFS strategy is equivalent to the standard weakly dominant strategy - bid one's valuation. But, as Cason and Plott (2014) show, only 16.7% of subjects bid their valuations, suggesting that very few recognize the dominant strategy. Intuitively, the problem is that almost all histories remain *payoff-relevant* (only Nature's draws at the two extremes of the distribution are payoff-irrelevant), so that selecting the optimal bid from many possible bids remains difficult.

Second, recent experiments studying contingent reasoning failures use clever designs to help pinpoint the source of contingent reasoning failures (Esponda and Vespa (2021), Martínez-Marquina, Niederle, and Vespa (2019)), providing insight into how the structure of the game affects choices in *strategically equivalent* games. Consider the auction game, AUCT, from Esponda and Vespa (2021) in which a subject participates in a single-player 'auction' where the highest bid wins and pays the bid of the second-highest bidder. The second bidder in the auction is represented by a computer bidder that is equally likely to bid \$0.50, \$4.50, or \$8.50. The subject chooses an integer bid from \$1 to \$8. She gets \$5.50 if she wins the auction and an outside value of \$3 if she does not. As can be seen in Figure 2.1, a single history is payoff-relevant: when the computer bids \$4.50. Bidding \$1-\$4 weakly dominates

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bidding any higher number and is WDFS. However, Esponda and Vespa (2021) find that 32.3% of all subjects choose a dominated bid between \$5-\$8.

Most interestingly, Esponda and Vespa (2021) find that the percentage of subjects choosing dominated bids drops to only 10.9% in a diagnostic treatment. The only difference in the diagnostic treatment is that subjects are asked to choose their bid *contingent on the payoff-relevant history,* $\mathbb{I}(I_i)$, in which the computer bids \$4.50 (they are told that computers' bids of \$8.50 and \$0.50 are payoff-irrelevant). Critically, subjects do not observe the computer's bid before making their choice in the diagnostic treatment so that the two games have identical strategy sets.[11]

The election game, ELECT, of Esponda and Vespa (2021) provides further evidence. In this game, the computer draws a white (w) or black (b) ball. There are two computerized voters. If the drawn ball is white (w), both computers vote white (WW). If the drawn ball is black (b), the computer voters randomly and independently vote for white or black. Without observing either the color of the drawn ball or the votes of the computers, a subject must choose between voting for Black (B) and voting for White (W). If the color chosen by the majority matches the color of the drawn ball, the subject gets \$5, and otherwise gets zero. As can be seen in Figure 2.1, two histories are payoff-relevant: when Nature plays b and the two computerized voters vote either WB or BW. Voting B dominates W and is WDFS. However, around 84.8% of all subjects choose the dominated bid, W. This fraction drops considerably to 46% in a diagnostic treatment where subjects are again asked to choose *contingent on (either of) the payoff-relevant histories,* $\mathbb{I}(I_i)$.

Importantly, the non-diagnostic versions of AUCT and ELECT, like BDM, satisfy *similar future actions and outcomes* (and *last mover*), suggesting something additional is necessary to make a game structure 'obvious'. Also, in both AUCT and its diagnostic version, bidding the dominant strategy is not obviously dominant, while in both ELECT and its diagnostic version, it is. Therefore, the differences in results between each game and its diagnostic version can not be explained by OD, and the failure rates are actually higher in the games in which the dominant strategy is OD. Again, this evidence suggests something about the structure of the game matters, as opposed to obvious dominance in terms of payoffs.[12]

[11]A literature studying sequential vs. simultaneous games finds that subjects do better in sequential versions of games (Carrillo and Palfrey (2011), Esponda and Vespa (2014), Magnani and Oprea (2017), Ngangoué and Weizsäcker (2021)). Sequential games are intuitively simpler because they focus subjects on a particular contingency. But, although this evidence is strongly suggestive, it does not directly speak to the structure of the game as sequential games are not strategically equivalent to their simultaneous counterparts.

[12]The evidence from Martínez-Marquina, Niederle, and Vespa (2019) also supports our notion of GSO. They show that subjects perform better in single-player auction games when uncertainty is removed, changing the game from one which does not satisfy our definition of salient contingency (defined below) to one that does. But, by removing uncertainty, their findings can also be explained

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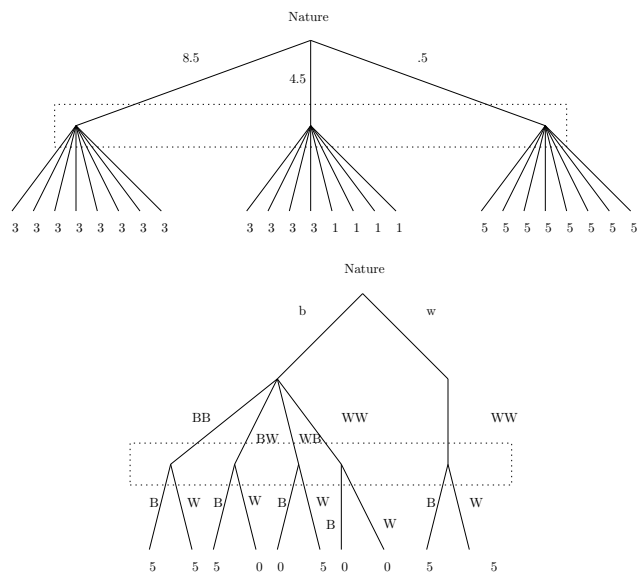

FIGURE 2.1. Extensive form representation of the AUCT (top) and ELECT (bottom) games from Esponda and Vespa (2021). The single player moves last in both games, bidding \$1-\$8 (from left to right) in AUCT and voting B or W in ELECT.

We use this evidence to motivate our final property, *salient contingency*. We will require GSO mechanisms to make it 'easy' for subjects to find the *payoff-relevant* histories (contingencies) in a way that depends solely on the game structure - *payoff-relevant* histories $\mathbb{I}(I_i)$ are salient because they are *pre-terminal*. Furthermore, best responding to any *pre-terminal* history guarantees a WDFS strategy.[13]

Definition. Subject i enjoys *salient contingency* in a game if each $I_i \in \mathcal{I}_i$ has $PT(I_i) \neq \emptyset$ and a WDFS, S_i^*, for type v_i can be constructed by selecting

$$S_i^*(I_i) \in \arg\max u_i(h, aS^{r1}, S_{-i}^1, v_i)$$

for any $h \in PT(I_i)$, $S_{-i}^1 \in \mathcal{S}_{-i}$, and $S^{r1} \in \mathcal{S}^r(I_i)$.

Note that BDM mechanisms, as well as the non-diagnostic versions of ELECT and AUCT in Esponda and Vespa (2021), do not satisfy *salient contingency*: all histories are *pre-terminal* but focusing on any arbitrary *pre-terminal* history does not lead to the dominant choice.

by the fact that the the dominant strategy is OD in the certain version only, providing another possible explanation.

[13]Practically speaking, making it easy to identify the payoff-relevant histories could be achieved in different ways. An experimentalist could nudge subjects in the right direction, make them play multiple practice rounds, or frame the experiment favorably. These aspects are all form part of the larger experimental design, but not part of the game itself.

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Finally, we can define a GSO mechanism:

Definition. We say that a mechanism is GSO and identifies all relevant types for player i if
(i) i is the *last mover*, and the game satisfies *similar future actions and outcomes*, *no cycles*, and *salient contingency*.
(ii) There are $|\mathcal{V}|$ strategies $\{S_j : j \in \mathcal{V}\}$ such that S_k is *weakly dominant against future selves* for type j if and only if $k = j$.
(iii) $\{S_j : j \in \mathcal{V}\}$ implements the social choice function.

Intuitively, in a GSO mechanism, a player need not understand exactly how she will play in each future contingency. Instead, she only needs to understand that she will do the same thing in all contingencies which look similar, and to best respond at the pre-terminal histories.

We know from Proposition 1 that a single payoff-relevant action cannot elicit more than two types. An elicitation mechanism (binary allocation mechanism) that elicits more than two types must then have more than one payoff-relevant action taken by i. We now show that such a mechanism cannot be both OSP and GSO. Li (2017) shows that all OSP binary allocation mechanisms must have the form of personal clock auctions in which some action can 'clinch' a certain outcome at every information set.

Proposition 3. *Take a game in which i is the last mover and has at least two payoff-relevant actions: there exist two information sets $I_i' \succ I_i$ that are reached by the identifying strategy, S_j, played by type v_j for some $S_{-i}^* \in \mathcal{S}_{-i}$. Suppose I_i' contains at least one payoff-relevant history. If the game satisfies similar future actions and outcomes, then it cannot be a personal clock auction, and vice-versa.*

Intuitively, taking a clinching action (which must exist at all information sets in a clock auction) in the first information set, I_i, renders all subsequent histories outcome-irrelevant. Additionally, similar future actions and outcomes extends such outcome-irrelevance to all subsequent histories that follow after any action taken at I_i. Thus, 'clock auction' and 'similar future actions and outcomes' together rule out the existence of a second information set with payoff-relevant histories. Thus a clock auction cannot be GSO, and a GSO mechanism cannot be a clock auction.

What, then, do binary allocation GSO mechanisms look like? Intuitively, they have an MPL-like structure in which subjects sequentially choose between an item and a varying price. In an MPL, a subject can focus on the possibility that each choice is drawn for payment when making the choice. But, *salient contingency* actually requires a specific, dynamic form of an MPL.

In Figure 2.2, we show the extensive form version of an MPL that is GSO (top left) and the standard way of implementing an MPL (top right), assuming that

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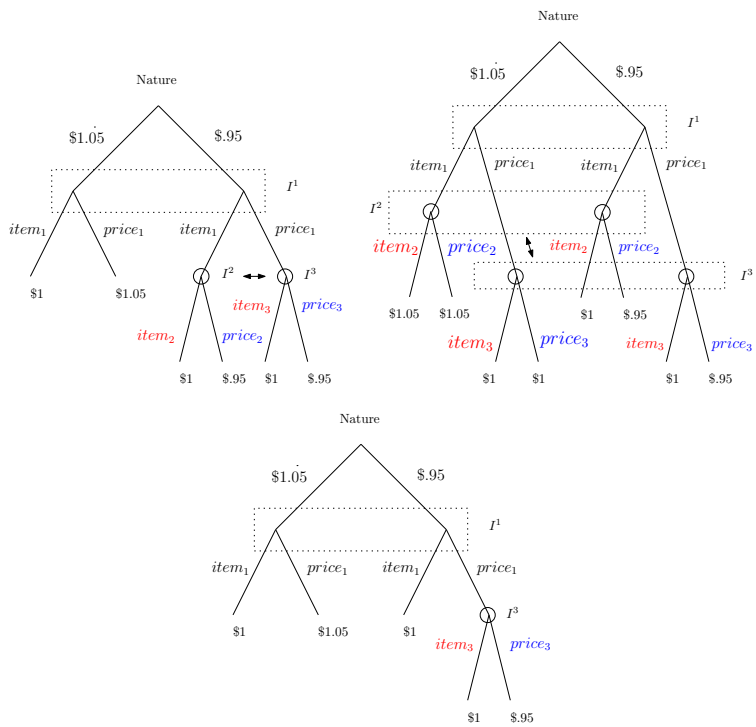

FIGURE 2.2. *Similar future actions and outcomes* in a GSO MPL (top left) and standard MPL (top right). Nature moves first to pick the price that 'counts', followed by player i who chooses between an item and a price at each information set. The information set and actions therein are numbered: for example, $item_3$ is the action that chooses the item in I^3. Similar information sets are marked with the $\leftrightarrow$ sign. Similar actions are marked using the same colors and font size. In the bottom panel, we present an OSP mechanism which is neither SOSP, nor satisfies *similar future actions and outcomes*.

subjects cannot change previously made choices. In each, Nature draws one of two dollar values (\$0.95 or \$1.05) as the price that determines the subject's payoff, and the participant i chooses between an item (\$1) and each of the prices (dollar values). For the MPL that is GSO, after information set I_i, a unique information set (circled) follows either action, so that the mapping among future information sets I^2 and I^3 is trivial. The game has a unique WDFS strategy of choosing the dollar amount at I^1 and the item thereafter. i also enjoys *salient contingency*: each *pre-terminal* history is *payoff-relevant*, and myopically best responding to it achieves the WDFS strategy. For the standard MPL on the right, I^2 and I^3 are similar, but both of the *pre-terminal* histories are *payoff-relevant* so that *salient*

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contingency does not hold. Finally, in the OSP game (bottom), choosing the item ends the game, and thus I^3 cannot be mapped to any information set.

3. Experimental Design

We designed our experiment with two main goals in mind. The first is to compare the performance of new elicitation mechanisms to the standard BDM mechanism. The second is to understand whether or not OSP mechanisms can fail because of future self-concerns.

3.1. **Main Treatments.** To achieve the first goal, we design three treatments, each consisting of a single task. Subjects are endowed with a virtual item that can be redeemed for \$1 or sold to the experimentalist, revealing a subject's willingness-to-accept (WTA) for \$1. Using an item of known value allows us to straightforwardly test the performance of each mechanism, as in Cason and Plott (2014).

Each of the three main treatments uses a different mechanism: BDM, OSP, and GSO. The BDM mechanism follows a standard implementation. Subjects were told to choose the value (WTA) between \$0 and \$1.50 at which they are willing to sell their \$1 item. The computer then randomly and uniformly drew a value in this interval. The subject kept the \$1 item if the random value was less than their WTA, and otherwise received the random value. The instructions were a computerized version of those in Plott and Zeiler (2005). Importantly, as in their instructions, subjects in our BDM treatment (only) were explicitly told that bidding \$1 was in their best interest, potentially improving their choices.

The OSP mechanism is a direct implementation of Li's (2017a) concept of obviously strategy-proof mechanisms in an individual decision-making setting. It utilizes a 'clock' price which begins at \$1.50 and decreases by \$0.05 each period until reaching \$0. In each period, a subject can either (i) attempt to sell their \$1 item at the given price, knowing that the computer had previously (uniformly) randomly chosen the price at which it would purchase the item, or (ii) choose to keep the item. The task ends when the computer purchases the item or the subject chooses to keep it. We chose a starting price of \$1.50 to reduce the amount of censoring that occurs in the mechanism; censoring occurs when the randomly chosen clock price stops the mechanism prior to the subject ever having chosen the item.[14]

Although several different GSO mechanisms are possible, for comparability purposes, we chose the GSO mechanism that is almost identical to the OSP mechanism,

[14]Ex post, an upper bound of \$1.50 does not seem to dramatically change behavior in BDM: the fraction of rational valuations we find is very similar to Cason and Plott (2014) who used an upper bound between \$4 and \$8 for a \$2 item.

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a dynamic MPL featuring a clock price that decreases each period and uses an identical set of prices.[15] In each period, subjects can choose either the price or the item. The task ends when the randomly chosen price is reached. Recall that the OSP mechanism is not GSO, and neither is the GSO mechanism OSP.

In both the GSO and OSP mechanisms, after making each choice, the payoff consequences of the choice were given and subjects were asked to confirm their choice. Unlike in the BDM treatment, in neither the OSP nor the GSO treatments were subjects informed of the optimal strategy: choosing the price at all clock prices above \$1 (or \$0.95) and then choosing the item in the remaining periods.

3.2. **Diagnostic Questions and Treatment.** The second goal of the experiment is to understand whether or not future self-concerns lead to failures of the OSP mechanism. If subjects don't recognize that they should choose the item at all clock prices less than \$0.95, then they may think that continuing to choose the clock price at high clock prices could lead to a payoff less than \$1, and therefore decide to choose the item too early. We hypothesized that such future self-concerns could lead to choosing the item at clock prices higher than \$1, and added two design features to the experiment to explore this possibility.

First, in the OSP and GSO treatments, we asked subjects a single incentivized question (\$0.50 for correct answer) after the main elicitation task. If the main task ended after the subject had chosen the item in some period, we reminded them of the first price at which they had chosen the item, and then asked them for the highest payoff they could have received if they had instead chosen the clock price that round. The purpose of this question was to determine whether or not subjects correctly understood the payoff consequences of their mistake.

Second, we added a treatment that uses a diagnostic SOSP elicitation mechanism. This elicitation scheme is identical to the OSP mechanism except that subjects were forced to choose the item choose the item when the price reaches \$1, a fact which we emphasized in the instructions. This mechanism is not a true elicitation mechanism due to the fact that the experimentalist must ex ante know the subject's valuation in order to stop her from choosing irrationally at lower prices. By forcing subjects to choose rationally at prices less than \$1, the mechanism becomes SOSP so that future self-concerns play no role. In the SOSP treatment, we include the same incentivized question as in the OSP and GSO treatments.

The OSP and GSO (and SOSP) mechanisms result in censoring of some subjects' valuations: if the randomly chosen clock price ends the mechanism before a subject has ever chosen the item, the subject's valuation is indeterminate. To address this issue, we also used the incentivized question to test for rationality. When the

[15]For example, an alternative GSO mechanism is one in which the clock price increases each period, keeping everything else the same.

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mechanism ends without the subject having chosen the item, we ask for the clock price at which the subject should first choose the item to maximize their bonus. In this way, we obtain measures of rationality for all subjects, avoiding the censoring issues inherent in the mechanisms.[16] But, because the question-based method and the dynamic mechanism may deliver different results, we analyze the data both with and without this additional information, as well as test for differences.

3.3. **Logistical Details.** All four treatments (BDM, OSP, GSO, and SOSP) were run on the Prolific platform in August and September of 2021.[17] We recruited between 297 and 301 subjects in each treatment. Average earnings were \$2.27, or about \$17.93/hour. Subjects first read instructions and then answered a series of non-incentivized quiz questions to assess their comprehension of the instructions. We did not prevent the participation of subjects who failed to answer the quiz questions correctly, nor include any attention check. The full set of instructions are reported in Appendix C.

4. Experimental Results

In Section 4.1, we summarize the main results and show that the OSP mechanism outperforms BDM. In Section 4.2, we show that many failures of the OSP mechanism are related to future self-concerns. Finally, in Section 4.3, we show that GSO outperforms OSP (and hence BDM), but is also imperfect.

4.1. **Summary of Results.** In our analysis, we focus on the subsamples of subjects that answered all quiz questions correctly in each treatment. Subjects that do not answer the questions correctly are likely to introduce noise as they are not paying attention to the instructions. To facilitate this comparison, we purposely made all four quiz questions in the dynamic treatments (OSP, GSO, and SOSP) identical to avoid differential selection across treatments. The BDM treatment had only three quiz questions which had to differ slightly from those in the dynamic treatments, but all of our conclusions about BDM continue to hold when we use the full samples instead of the filtered samples.

Although the dynamic mechanisms (OSP, GSO, and SOSP) elicit truthful individual valuations, if one pools across subjects, we obtain biased estimates of the *distribution* of valuations due to the fact that subjects who switch earlier are less likely to be censored. We account for this bias in two ways. First, we create a 'trimmed' sample of subjects that includes only the subsample of subjects for which the randomly chosen clock price was less than \$1, ensuring rational valuations are

[16] Of course, asking for the payoff-maximizing price at which to switch is only possible in a diagnostic environment where this price is known.

[17] Because of gender imbalances in Prolific's subject pool, we recruited male and female subjects separately.

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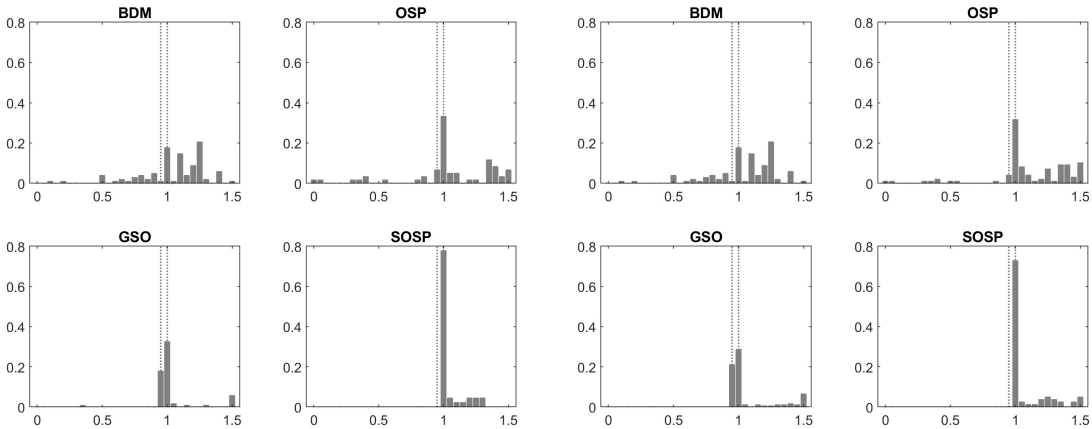
FIGURE 4.1. Filtered Valuations



Notes: Histograms (proportions) of valuations in all treatments for those subjects that answered all quiz questions correctly. The dashed vertical lines indicate rational valuations ($0.95 and $1). The left four panels are for the trimmed samples of subjects (those for which the randomly chosen clock price was less than $1) and the right four panels are for all subjects (using valuations elicited via the incentivized question). The SOSP mechanism is a diagnostic mechanism that prevents valuations below $1.

not censored. In this sample, we also include all subjects that were censored at valuations less than $1, taking the last observed price as a subject's valuation if they were censored. In this way, we correctly observe the proportion of subjects with high (>$1), rational ($1 or $0.95, due to indifference at $1), and low (<$1) valuations.[18] In the second, 'full', sample, we include all subjects, utilizing the valuations obtained through the incentivized question for censored subjects.

Figure 4.1 provides histograms of the elicited valuations in all four treatments, filtered on subjects that answered all quiz questions correctly.[19] The left four panels are for the trimmed samples and the right are for the full samples. For the GSO mechanism, we exclude subjects that switched more than once between the price and the item because their valuation is then indeterminate.[20] The corresponding plots for all subjects (not filtering on the quiz questions) look similar, and are provided in Appendix A.

In Figure 4.1, we observe clear variation across mechanisms but little difference between the left and right panels, indicating that including all of the data may

[18]The actual valuations for subjects with valuations less than $1 will be overestimated in some cases, but, in practice, there are very few such observations.

[19]The bins are $0.05 wide so that for the dynamic mechanisms none of the valuations are pooled.

[20]Almost every subject that switched more than once first does so at a price greater than $1, so that if we take the first switch as their valuation, the fraction of subjects making rational choices is almost identical.

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TABLE 1. Summary of Valuations

Treatment	Sample	Filtered on Quiz	N	Rational Valuations	Valuations >$1	Valuations <$0.95	Multiple Switch
BDM	all	yes	102	15.7%	60.8%	23.5%	-
	all	no	297	19.9%	62.0%	18.2%	-
OSP	trimmed	yes	60	40.0%	43.3%	16.7%	-
	all	yes	98	35.7%	55.1%	9.2%	-
	trimmed	no	193	31.1%	57.5%	11.4%	-
	all	no	300	28.7%	66.0%	5.3%	-
GSO	trimmed	yes	123	50.4%	8.9%	0.8%	39.8%
	all	yes	185	49.7%	14.6%	0%	35.7%
	trimmed	no	195	40.0%	7.7%	1.0%	51.3%
	all	no	299	40.5%	17.4%	0.7%	41.5%
SOSP	trimmed	yes	45	77.8%	22.2%	-	-
	all	yes	81	72.8%	27.2%	-	-
	trimmed	no	186	50.0%	50.0%	-	-
	all	no	301	44.2%	55.8%	-	-

Notes: Fraction of subjects with rational valuations and those above and below $1 for each treatment. For the OSP, GSO, and SOSP treatments, results for the trimmed sample of subjects (those for which the randomly chosen clock price was less than $1) and for all subjects (using valuations elicited via the incentivized question) are reported. For all mechanisms, results are also filtered to include only the sample of subjects that answered all quiz questions correctly. The SOSP mechanism is a diagnostic mechanism that prevents valuations below $1.

be reasonable. To perform statistical comparisons, we begin by summarizing the data in Table 1. We separate valuations into rational ($1 and $0.95), as well as valuations above $1 and below $0.95. For GSO, we also report the frequency of multiple-switching behavior. We report the data using the filtered samples corresponding to Figure 4.1 but, for completeness, also report the raw, unfiltered data. N indicates the size of each sample. With the exception of the BDM treatment, we see considerably more rational behavior in the filtered samples, consistent with those not answering the quiz questions correctly not paying attention.[21]

We first ask whether or not the incentivized question that asked for the payoff-maximizing price produces similar results to the dynamic mechanisms, so that we can test for rationality while avoiding censoring issues. We cannot compare the fraction of rational valuations elicited using choices from the dynamic mechanism to those elicited via the question because those elicited via the question are selected based on not having already chosen the item. So, instead, we compare the fraction

[21]The fact that understanding the quiz questions leads to better decisions in the OSP, GSO, and SOSP mechanisms, and not in the BDM mechanism is consistent with the findings of Danz, Vesterlund, and Wilson (2020). They find that providing detailed instructions for a belief elicitation task actually *decreases* the fraction of subjects that report correct probabilities. For non-transparent elicitation mechanisms such as BDM and the binarized scoring rule used in Danz, Vesterlund, and Wilson (2020), it seems that understanding the detailed instructions required to implement the mechanism does not improve the mechanism's performance, and may even worsen it.

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of rational valuations for the subsample of subjects that were censored at prices greater than or equal to \$1 to that for those censored below \$1 (the trimmed sample) using a chi-squared test. We find no difference between these two samples in any of the three treatments: OSP ($pval = 0.454$), GSO ($pval = 0.687$), or SOSP ($pval = 0.497$). Furthermore, the point estimates are very similar across the samples (although slightly higher for the valuations elicited via the mechanism directly): 28.9% vs 35.7% in OSP, 46.7% vs. 49.7% in GSO, and 66.7% vs. 72.8% in SOSP. Given these results, we are quite comfortable in using the full sample of subjects, but we continue to report results for both this sample and the trimmed sample.

We begin our assessment of the performance of the mechanisms by comparing BDM to OSP, expecting OSP to do better due to the fact that reporting a valuation of \$1 (switching at \$1) is SOD in OSP, but not in BDM. To make this comparison, we perform a chi-squared test of the relative proportions of rational valuations. In doing so, we find that OSP outperforms BDM whether we include all of the data or only the trimmed sample ($pval < 0.001$). The increases in rational decisions are also large in magnitude, going from 15.7% in BDM to between 35.7% and 40% in OSP, depending on the sample size.

Result 1: *The OSP mechanism greatly outperforms BDM in eliciting truthful valuations.*

The poor performance of BDM is not unexpected given previous findings. In the closest related studies, Cason and Plott (2014) find 16.7% of inexperienced subjects' valuations are rational, while Martin, Munoz, and Rodriguez (2020) find 17.4% in their replication of the previous study. Both of these results are similar to the 15.7% we find.[22]

The improved performance of the OSP mechanism provides additional evidence that OSP mechanisms in general outperform their more traditional counterparts (as in Li (2017a). However, the fact that only 40% of subjects correctly reveal a valuation of \$1 shows that its performance is far from perfect. We study these failures in more detail in the following section.

4.2. **Future Self-Concerns.** Our main question is whether or not failures of the OSP mechanism are due to future self-concerns: subjects may choose the item at clock prices greater than \$1 to avoid future mistakes that lead to payoffs less than \$1. We test this hypothesis by comparing behavior in the OSP and SOSP mechanisms.

[22]Previous studies treated \$0.95-\$1.05 as rational instead of \$0.95-\$1.00. If we use the wider range, 19.6% of bids would be rational, which is slightly higher than in the previous studies, perhaps because we explicitly told subjects that bidding \$1 was in their best interest.

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Recall that the SOSP diagnostic mechanism is identical to the OSP mechanism except that subjects were forced to keep the item when the price reached \$1, thus preventing them from ever earning less than \$1. If future self-concerns are driving mistakes in the OSP mechanism, we would expect more valuations greater than \$1 in OSP than in SOSP. Using chi-squared tests, we indeed find a higher number of high valuations in OSP both in the trimmed sample ($pval = 0.024$) and the full sample ($pval < 0.001$). Furthermore, the fraction of valuations greater than \$1 is much higher in OSP than in SOSP, about 25 percentage points across samples.

We are further interested in whether or not subjects with switch points greater than \$1 in the OSP mechanism understand that choosing \$1 means they forgo the possibility of earning more than \$1, or are instead simply acting noisily. The fraction of subjects that choose the item at a clock price greater \$1 *and* correctly report the payoff consequences of doing so is 32% in the full sample and 38% in the trimmed sample. In the SOSP treatment, the corresponding fractions are only 14% and 13%, differences that are highly significant ($pval < 0.005$). Thus, it appears that not only do subjects choose the item more often at high valuations in the OSP treatment compared to the diagnostic SOSP treatment, they do so understanding that they are sacrificing the chance to earn more then \$1, perhaps because future self-concerns lead them to worry that they may earn less than \$1 by continuing to choose the price.

These results provide the first empirical evidence (of which we're aware) of a potential limitation of dynamic mechanisms, including OSP mechanisms: subjects do not appear to correctly forecast their optimal future actions.

Result 2: *Subjects make mistakes - choose the item at clock prices greater than one dollar - more often in a mechanism (OSP) which requires them to correctly forecast their future actions than in a diagnostic mechanism that does not (SOSP). Furthermore, they do so understanding that they may leave money on the table.*

4.3. **GSO Results.** Given that the OSP mechanism suffers from future self-concerns, we turn to the question of whether or not the GSO mechanism does better. Comparing the dynamic mechanisms using a chi-squared test, we find that GSO outperforms OSP in the full sample ($pval = 0.024$) but not in the trimmed sample ($pval = 0.186$). The magnitude of the improvements are large, improving from 35-40% rational in OSP to about 50% in GSO, so we attribute the lack of statistical significance in the trimmed sample as being due to small sample size after filtering both on the quiz questions and the ending price.

Result 3: *The GSO mechanism modestly outperforms the OSP mechanism (and therefore greatly outperforms BDM) in eliciting truthful valuations.*

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Future self-concerns cannot play a role in the GSO mechanism, so we also expect more valuations above \$1 in OSP relative to GSO. Indeed, in both samples we find significantly less high valuations in GSO than in OSP ($pval < 0.001$), and significantly less high valuations in which subjects correctly report the payoff consequences ($pval < 0.001$). We also find significantly less valuations less than \$1 in GSO than in OSP in both samples ($pval < 0.001$).

Given significantly fewer high and low valuations, we would expect GSO to outperform OSP more than it does. However, GSO allows for a form of behavioral mistake that OSP does not - switching between the price and the item in multiple periods. As in the large literature using multiple price lists for elicitation, we find multiple switching to be common, with 35-40% of subjects switching multiple times. In fact, multiple switching is by far the most common form of failure of the GSO mechanism.

5. Discussion

We have used elicitation schemes as a testing ground for the study of mechanisms robust to bounded rationality, both theoretically and empirically. We focus on future self-proof mechanisms that do not require subjects to understand their future actions. Theoretically, we first show that it is impossible to design an elicitation scheme that is strongly obviously strategy-proof: one that does not require reasoning about one's future actions and also maintains the property in which the best payoff from the truth-telling strategy exceeds that of any deviation. Given this result, we introduce the concept of *game-structure obviousness*: a game has an obvious structure if subjects can understand the payoff consequences of each of their present actions without understanding their other actions or the actions of others, and it is easy to identify the dominant strategy.

Empirically, we provide evidence that OSP and GSO elicitation mechanisms outperform standard BDM, but that future self-concerns limit the OSP mechanism's ability to elicit truthful valuations. From a purely practical point of view, our results are important because they provide a pair of simple-to-use mechanisms that outperform BDM. Perhaps more importantly though, they suggest that the FSP property may be important in designing robust mechanisms more generally. Our FSP (GSO) mechanism performs best, but is not a complete panacea as it leads to mistakes in which subjects make different decisions when faced with very similar choices.

From the comparison between the OSP and GSO results, it may be tempting to conclude that a single source of irrationality is responsible for failures in both mechanisms. Indeed, almost all subjects that switch multiple times in the GSO mechanism first choose the item at a price greater than \$1, so that if these first

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switches were taken to be their valuations, the frequencies of high valuations in the OSP and GSO mechanisms would line up quite well.[23] However, if the mistakes did stem from the same source, it would not explain why high valuations decrease substantially in the SOSP diagnostic mechanism where the need for a subject to understand his or her future actions is removed.

Instead, we believe the empirical results point to a fundamental tradeoff in the design of elicitation mechanisms, and perhaps robust mechanism design more broadly. The OSP property requires moving from a static mechanism to a dynamic mechanism that intuitively makes it simpler to understand each choice. But, the dynamic nature of OSP mechanisms then requires rational foresight of one's future actions, which some subjects do not appear to have. The benefit of future self-proof mechanisms, achieved via the GSO property, is that it eliminates the need for rational foresight by creating a game structure in which similar decisions are made at each information set. Unfortunately, the benefit of this structure is also its weakness - subjects seem averse to making consistent decisions when all decisions are similar. Although we can't be certain why this behavior occurs, a recent paper by Agranov, Healy, and Nielsen (2021) suggests that it may be rooted in a preference for randomization. They show that even in a setting in which subjects sequentially make multiple *identical* choices between a safe and risky option, they do not choose consistently. On the other hand, recent work with multiple price lists has shown that nudges can reduce multiple switching behavior (Yu, Zhang, and Zuo (2021)), suggesting that the behavior may not be a result of immutable preferences. It seems that a necessary step to designing more robust mechanisms is to get at the root of inconsistent choices among similar, even identical, choice sets.

Motivated by theoretical considerations and experimental evidence from papers studying contingent reasoning, we adopted a strong definition of GSO that leads to an elicitation mechanism with a dynamic MPL structure. Furthermore, to facilitate comparison, we chose the GSO mechanism most similar to the OSP mechanism, which requires a decreasing clock price. Future empirical work could compare different GSO mechanisms and a standard MPL (which is not GSO), exploring different aspects of the experimental design (instructions, experience, nudges to prevent multiple switching behavior, etc.) in an attempt to identify the 'best' practical mechanism. Future work should also study GSO mechanisms for the elicitation of lotteries and subjective probabilities.

[23]For example, one could posit a simple model in which subjects generally choose the dominant action, but sometimes 'tremble'. This model would lead to high valuations in the OSP mechanism and multiple switches in the OSP mechanism.

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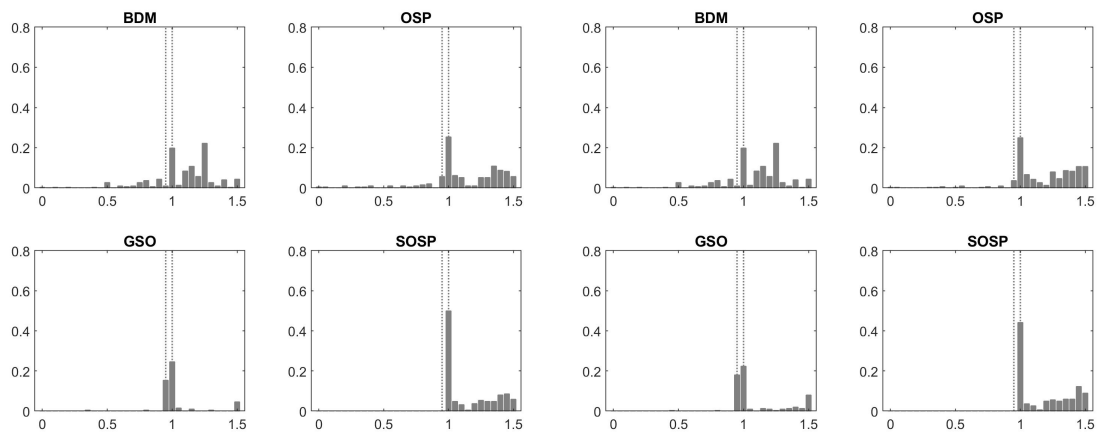
Appendix A: Additional Results

Figure 5.1. Unfiltered Valuations



Notes: Histograms (proportions) of valuations in all treatments for all subjects. The dashed vertical lines indicate rational valuations ($0.95 and $1). The left four panels are for the trimmed samples of subjects (those for which the randomly chosen clock price was less than $1) and the right four panels are for all subjects (using valuations elicited via the incentivized question).

Appendix B: Omitted Proofs

Omitted Proofs.

Proof. (Proposition 1). Part i: Let the set of possible payoffs from action a_i be given by $\Pi_{a_i}(v) \subseteq \Pi(v) \equiv \{\pi(v) : \pi(v) = pv + r, p \in \{0, 1, 2.., m\}, r \in \mathbb{R}\}$.[24] Suppose such an identifying SOSP mechanism exists. Then, there must exist a unique action a_i that is SOD at every valuation v_i. We will show a contradiction in three steps.

Step 1: Let there exist some type $i \in \mathcal{V}$, for whom the identifying action a_i has $\Pi_i(v)$ with more than one value. We change the outcomes promised from a_i such that $\Pi_i(v)$ becomes

$$\Pi'_{a_i}(v) = \arg\max_{\pi \in \Pi_{a_i}} \pi(v_i)$$

If $\arg\max_{\pi \in \Pi_{a_i}} \pi(v_i)$ has multiple solutions, choose any one randomly. For example, if the action a_i paid (1 item + $10) or ($20) with equal chance was used to identify a type with valuation $v =$$15, then, this step would modify the action to now pay (1 item + $10) for certain instead. We claim that this new set of offered

[24] For example, if action a_i pays (1 item + $10) or ($20) with equal chance, then, $\Pi_i^m(v) = \{v + 10, 20\}$. If action a_i pays (1 item + $10) for certain, then, $\Pi_i^m(v) = \{v + 10\}$.

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actions creates a new SOSP mechanism that still identifies all the types. For type i, action a_i is still SOD:

$$\min \Pi'_{a_i}(v_i) \quad \geq \quad \underbrace{\min \Pi_{a_i}(v_i) \geq \max \Pi_{a_j}(v_i)}_{\text{because the action was previously SOSP}} \qquad \forall j \neq i$$

and action $k \neq i$ stays not SOD for type i, as $\min \Pi_{a_k}(v_i)$ and $\max \Pi_{a_l}(v_i)$ are unchanged. For all other types the proof follows similarly.

Step 2: Applying Step 1 recursively for each type creates a new SOSP mechanism where $\Pi_{a_i}(v)$ is a singleton for every identifying action a_i. (Thus each action now pays a certain amount of items and money.)

Step 3: Given Step 2 and $m < n$, there must exist at least two types $x, y \in \mathbb{N}$ such that their identifying actions, a_x and a_y, allocate an identical number of items. Let the payoffs from those identifying actions be $\Pi_x(v) = pv + r_x$ and $\Pi_y = pv + r_y$. Assume $r_x > r_y$. Then, both types would take action a_x because $px + r_x > px + r_y$. If $r_x < r_y$, then both take action a_y. If $r_x = r_y$,both types are indifferent between actions, so that the types are not strictly identified. Thus, in all three cases, we have a contradiction.

Part ii: Pick n actions $(a_i)_{i=1}^n$ such that $\Pi_{a_n}(v) = (n-1)v$ and $\Pi_{a_i}(v) = (i-1)v + \sum_{j=i}^{n-1} \frac{v_{j+1}+v_j}{2}$ for $i \in \{1, ..n-1\}$. Action a_i identifies (v_i). We skip the calculations for brevity. □

Proof. (Proposition 2) Step 1: When $\mathbb{I}(I_i)$ is empty, all actions lead to the same payoffs, so all actions are equally good and any of them can be used to construct a WDFS strategy at that information set. The crux of the proof lies in showing that T is non-empty when $\mathbb{I}(I_i)$ is non-empty. We use induction on the size of $\mathcal{A}(I_i)$.
If $|\mathcal{A}(I_i)| = 1$, $T = \mathcal{A}(I_i)$. Suppose, $|\mathcal{A}(I_i)| = 2$ with actions a and a'. If $u_i(h, aS_i^r, S_{-i}, v_i) = u_i(h, a'S_i^r, S_{-i}, v_i) \;\; \forall h, S_i^r, S_{-i}$, then $T = \{a, a'\}$. Otherwise, there exists, $h^1 \in I_i$, $S_{-i}^1 \in \mathcal{S}_{-i}$ and $S_i^{r1} \in \mathcal{S}^r(I_i)$ such that,

$$u_i(h^1, aS_i^{r1}, S_{-i}^1, v_i) \quad > \quad u_i(h^1, a'S_i^{r1}, S_{-i}^1, v_i)$$

[25]so that identical future payoffs implies

$$u_i(h, aS_i^r, S_{-i}, v_i) \quad \geq \quad u_i(h, a'S_i^r, S_{-i}, v_i) \;\; \forall h, S_i^r, S_{-i}$$

and thus, $T = \{a\}$.
Suppose, T is non-empty for all $|\mathcal{A}(I_i)| \leq n$. Now, for some $|\mathcal{A}(I_i)| = n+1$, take any $\hat{a} \in \mathcal{A}(I_i)$. Thus, there exists $a^* \in \mathcal{A}(I_i) \backslash \{a\}$ such that

$$u_i(h, a^*S^{r1}, S_{-i}^1, v_i) \geq u_i(h, a'S^{r1}, S_{-i}^1, v_i) \;\; \forall a' \in \mathcal{A}(I_i) \backslash \{\hat{a}\}, h \in I_i, S_{-i}^1, S^{r1}$$

[25]We have chosen a to be better than a' without loss of generality.

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Now, comparing a^* with $\hat{a}$, we get $a^* \in T$ if

$$u_i(h, a^* S^{r1}, S^1_{-i}, v_i) \geq u_i^G(h, aS^{r1}, S^1_{-i}, v_i) \quad \forall h \in I_i, S^1_{-i}, S^{r1}$$

else, by the *no cycles* condition,

$$u_i(h, \hat{a}S^{r1}, S^1_{-i}, v_i) \geq u_i^G(h, aS^{r1}, S^1_{-i}, v_i) \quad \forall h \in I_i, S^1_{-i}, S^{r1}$$

Step 2: Let $a \in T$. By definition, $\forall h \in I_i$ $\forall S_{-i} \in \mathcal{S}_{-i}$ and $\forall S_i^r \in \mathcal{S}^r(I_i)$,

$$u_i(h, aS_i^r, S_{-i}, v_i) \quad \geq \quad u_i(h, a' S_i^r, S_{-i}, v_i) \quad \forall a' \in \mathcal{A}(I_i)$$

Thus, $S_i^*(I_i) = a$ constructs a WDFS strategy at I_i. □

Proof. (Proposition 3) Suppose the game satisfies similar future actions and outcomes, and it is a personal clock auction. We will show a contradiction.
$I_i' \succ I_i$ and perfect recall together imply that for any $h_1 \in I_i'$ there must exist $h_2 \in I_i$ such that $h_1 \succ h_2$. Using this property, let $h \in I_i$ be the precedent to the payoff-relevant history $h' \in I_i'$. Thus, $h' \succ h$.
Payoff-relevance at the later history $h' \in I_i'$ implies that the second condition of *similar future outcomes and payoffs* cannot hold, and hence, there must exist two actions $b_1, b_2 \in \mathcal{A}(I_i')$ and a I_i'-restricted strategy $S^{r*} \in \mathcal{S}^r(I_i')$ such that,

$$\begin{aligned} & u(h', b_1 S^{r*}, S^*_{-i}, v_j) \neq u(h', b_2 S^{r*}, S^*_{-i}, v_j) \\ \implies & z_i(h', b_1 S^{r*}, S^*_{-i}) \neq z_i(h', b_2 S^{r*}, S^*_{-i}) \end{aligned} \tag{5.1}$$

At the earlier history, h, the similarity of future actions and outcomes implies either

$$\underbrace{z_i(h, aS^{r1}, S^*_{-i}) = z_i(h, aS^{r2}, S^*_{-i}) \quad \forall a \in \mathcal{A}(I_i),\ \forall S^{r1}, S^{r2} \in \mathcal{S}^r(I_i)}_{\text{future actions are irrelevant to the outcome}} \tag{5.2}$$

or

$$\underbrace{z_i(h, a_1 S^{r1}, S^*_{-i}) = z_i(h, a_2 S^{r1}, S^*_{-i}) \quad \forall a_1, a_2 \in \mathcal{A}(I_i),\ \forall S^{r1}, S^{r2} \in \mathcal{S}^r(I_i)}_{\text{outcome under any future action is unaffected by current actions}} \tag{5.3}$$

must hold.
By the definition of a personal clock auction, $\mathcal{A}(I_i)$ must include a quitting action a^q such if i plays a^q, then at every subsequent outcome of the game, she keeps the item and gets a constant amount of money $\bar{t}_i$. We denote this outcome as $\bar{z}_i = (1, \bar{t}_i)$. Let $a^* \in \mathcal{A}(I_i)$ be the action at $h \in I_i$ that leads from h to h'. Pick $S^{r1} \in \mathcal{S}^r(I_i)$ such that (i) the game reaches h' from h[26], and b_1 is played at h', and $S^{r1} = S^{r*}$ at all information sets $I \succ I_i'$ that succeed I_i'. Pick $S^{r2} \in \mathcal{S}^r(I_i)$ such that (i) the game reaches h' from h, and b_2 is played at h', and $S^{r2} = S^{r*}$ at all information sets $I \succ I_i'$.

[26]As i is the last mover, and $h' \succ h$, this can always be guaranteed.

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If (5.2) holds, then using $a = a^*$,

$$\begin{aligned} & z_i(h, a^* S^{r1}, S^*_{-i}) = z_i(h, a^* S^{r2}, S^*_{-i}) \\ \implies & z_i(h', b_1 S^{r*}, S^*_{-i}) = z_i(h', b_2 S^{r*}, S^*_{-i}) \end{aligned}$$

creates a contradiction to (5.1). The implication follows by the construction of S^{r1}, S^{r2}.

If (5.3) holds, then,

$$\begin{aligned} & z_i(h, a^* S^{rk}, S^*_{-i}) = z_i(h, a^q S^{rk}, S^*_{-i}) = \bar{z}_i \ \forall k = 1, 2 \\ \iff & z_i(h', b_1 S^{r*}, S^{1*}_{-i}) = z_i(h', b_2 S^{r*}, S^*_{-i}) = \bar{z}_i \end{aligned}$$

creates a contradiction to (5.1). The implication follows by the construction of S^{r1}, S^{r2} and by $z(h, a^q S^{rk}, S^*_{-i}) = \bar{z}$ for $k = 1, 2$. □

Appendix C: Instructions

We include the instructions and screenshots of the main task and incentivized questions for the BDM, OSP, and GSO treatments, in that order. The instructions for the SOSP treatment are identical to those for the OSP treatment except for a single sentence, so we include only the different page.

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USC University of Southern California

Instructions

In this experiment, you will complete a single task. You own an item that can be redeemed for a bonus of $1.00. The experimenter wants to buy it. Your task is to make an offer for the item.

Your offer will be compared to an unrelated fixed offer that is equally likely to be any number between $0.00 and $1.50. If your offer is less than or the same as the fixed offer, then you sell the item. You had the low offer, so you are the seller. But, here's the interesting part. You do not receive your offer. Instead, you receive the fixed offer, a price higher than your offer.

Example: if you offer $0.75 and the fixed offer is $0.80, you have the low offer. You sell the item and you receive the fixed offer of $0.80.

If your offer is more than the fixed offer then you do not sell your item. You keep the item.

Example: if you offer $1.25 and the fixed offer is $1.20, you do not have the low offer. Therefore, you do not sell the item.

You **should offer the minimum amount you would be willing to accept** in exchange for the item you own. There are no advantages to strategic behavior in this task. Your best strategy is to determine your personal value for the item and offer that value.

Why is my best strategy to bid the minimum I'd be willing to accept? Let's go back to the example:

Say I decide that the minimum I'd be willing to accept for my item is $1.25. What happens if I ask for less than $1.25? Say I ask for only $1.10. If the fixed amount is, say, $1.15, then I have to sell my item. I lose out because I have to give up the item which I think is worth $1.25, but I only get $1.15 in exchange.

What happens if I ask for more than $1.25? Say I ask for $1.50. If the fixed amount is $1.40, then I do not sell. But, had I bid $1.25, I would have sold the item and received $1.40 for an item that I think is worth only $1.25. I would have done better.

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When you have understood these instructions, please proceed to the next page to answer several comprehension questions.

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Comprehension Questions

Please answer the following comprehension questions. Click 'Show/hide instructions' if you need to review the instructions again.

Show/hide instructions

1. Please indicate which of the following statements is true:

the fixed offer is more likely to be $1.50 than $1.00

the fixed offer is more likely to be $1.00 than $1.50

the fixed offer is equally like to be $1.00 or $1.50

2. What will be your bonus if the fixed offer is $1.25 and your offer is $1.15?

$1.25

the value of the item

$1.15

3. What will be your bonus if the fixed offer is $1.15 and your offer is $1.25?

$1.25

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the value of the item

$1.15

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Main Task

Your item is a card that can be redeemed for a bonus of $1.00. You may sell it to the experimenter.

Please enter your offer as a number between 0.00 and 1.50 (no dollar sign) in the box below and then click the right arrow to confirm.

Show/hide instructions

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Instructions

In this experiment, you will complete a single main task which we describe below. In addition, we will ask you one paid question after the main task. If you answer it correctly, we will give you an additional bonus of $0.50. This paid question is separate from the comprehension questions which follow these instructions.

INSTRUCTIONS FOR MAIN TASK:

Period: 1
Price: $1.50

Please choose either the left or right button.

Try to Sell **Exit the Market**

In this task, you have a virtual item which is worth $1.00.

You can try to sell your item to the computer by participating in a market where you will make choices over several periods. Each period, you will click either Try to Sell or Exit the Market, and then confirm your choice.

If you choose Exit the Market in any period, the task ends. You keep the item and can no longer sell it. In this case, your bonus is $1.00.

If you choose Try to Sell, you offer your item for sale to the computer at the current price (shown just below the current period). The computer has already randomly selected the price at which it will buy the item from you. If it buys the item from you, you receive the current price. If not, we go to the next period, the price decreases by $0.05, and you again choose between Try to Sell and Exit the Market.

For example, in the picture above, you are in Period 1 and the current price is $1.50. if you click on Try to Sell, you would get $1.50 if the computer has chosen to buy the item from you at $1.50 (and otherwise you will go to the next period). If you instead click on Exit the Market, you would get the value of the item ($1.00) for sure.

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Each price between $1.50 and $0.00 is equally likely to be the price at which the computer decided to buy the item from you. In the first period, the price begins at $1.50. It decreases by $0.05 after each period that the computer does not buy the item. The task ends when the price decreases to $0.00 or you choose Exit the Market, whichever comes first.

When you have understood these instructions, please proceed to the next page to answer several comprehension questions.

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Comprehension Questions

Please answer the following comprehension questions. Click 'Show/hide instructions' if you need to review the instructions again.

Show/hide instructions

1. Please indicate which of the following statements is true:

the price decreases each period

the value of the fixed item does not change

both of the above are true

2. What will be your bonus if you choose Try to Sell when the price is $1.25 and the computer has chosen to buy the item from you at that price?

$1.25

the value of the item

it is random

3. What will be your bonus if you choose Exit the Market when the price is $1.25?

$1.25

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the value of the item

it is random

4. What will happen if you choose Try to Sell when the price is $1.25 and the computer does not buy the item from you at that price?

task continues, you can Try to Sell at $1.20 or Exit the Market in the next period

task ends, you earn a bonus of $1.25

task ends, you earn the value of the item as a bonus

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Main Task

Period: 1
Price: $1.50

Please choose either the left or right button.

Try to Sell

Exit the Market

Show/hide instructions

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Paid Question

Please answer the following question. You will receive an additional bonus of $0.50 for a correct answer.

Show/hide instructions

In the main task you just completed, you always chose Try to Sell. What is the **first** dollar amount at which you should choose the Exit the Market to maximize your bonus?

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Paid Question

Please answer the following question. You will receive an additional bonus of $0.50 for a correct answer.

Show/hide instructions

In the main task you just completed, you first chose the Exit the Market when the price was $1.35. If you had instead chosen Try to Sell, what is the maximum bonus you could have received in this period?

$1.35

the value of the item

less than $1.35

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Instructions

In this experiment, you will complete a single main task which we describe below. In addition, we will ask you one paid question after the main task. If you answer it correctly, we will give you an additional bonus of $0.50. This paid question is separate from the comprehension questions which follow these instructions.

INSTRUCTIONS FOR MAIN TASK:

Please choose either the left or right button. Each button indicates your bonus if the computer chose this period to end the task. If not, we go to the next period.

$1.50 **$1.00**

In this task, you will make choices over several periods. Each period, you will click to choose the dollar amount on the left, or the fixed item on the right ($1.00), and then confirm your choice. The computer has already randomly selected the period in which the task will end. Your bonus will depend **only on the choice you make in the period that the computer selected to end the task.**

For example, in the picture above, if the computer selected this period to end the task, your bonus will depend on which button you click. If you click on $1.50, you would get $1.50. If you instead click on the item on the right, you would get the value of the item. If the computer hasn't chosen this period to end the task, you go to the next period, and choose between $1.45 and the item.

Each period is equally likely to be the period that the computer decided to end the task. The dollar amount on the left begins at $1.50, decreases by $0.05 each period, and the task will end before it decreases to $0.00. The item on the right does not change.

When you have understood these instructions, please proceed to the next page to answer several comprehension questions.

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Comprehension Questions

Please answer the following comprehension questions. Click 'Show/hide instructions' if you need to review the instructions again.

Show/hide instructions

1. Please indicate which of the following statements is true:

the dollar amount on the left decreases each period

the fixed item on the right never changes

both of the above are true

2. What will be your bonus if the computer ends the task when you choose the dollar amount of $1.25?

$1.25

the value of the item

it is random

3. What will be your bonus if the computer ends the task when you choose the fixed item on the right?

$1.25

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the value of the item

it is random

4. What will happen if you choose the dollar amount of $1.25 instead of the item and the computer **does not** end the task?

task continues, you choose between $1.20 and the item in the next period

task ends, you earn bonus of $1.25

task ends, you earn the item on the right as bonus

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Main Task

Period: 1

Please choose either the left or right button. Each button indicates your bonus if the computer chose this period to end the task. If not, we go to the next period.

$1.50 **$1.00**

Show/hide instructions

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Paid Questions

Please answer the following question. You will receive an additional bonus of $0.50 for a correct answer.

Show/hide instructions

In the main task you just completed, you always chose the dollar amount on the left. What is the **first** dollar amount at which you should choose the item on the right to maximize your bonus?

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Paid Questions

Please answer the following question. You will receive an additional bonus of $0.50 for a correct answer.

Show/hide instructions

In the main task you just completed, you first chose the item on the right when the dollar amount on the left was $1.40. If you had instead chosen the dollar amount, what is the maximum bonus you could have received in this period?

$1.40

the value of the item

less than $1.40

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Instructions

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If you choose Exit the Market in any period, the task ends. You keep the item and can no longer sell it. In this case, your bonus is $1.00.

If you choose Try to Sell, you offer your item for sale to the computer at the current price (shown just below the current period). The computer has already randomly selected the price at which it will buy the item from you. If it buys the item from you, you receive the current price. If not, we go to the next period, the price decreases by $0.05, and you again choose between Try to Sell and Exit the Market.

You will be forced to choose Exit the Market when the price reaches $1.00. Therefore, you cannot earn a bonus less than $1.00.

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For example, in the picture above, you are in Period 1 and the current price is $1.50. if you click on Try to Sell, you would get $1.50 if the computer has chosen to buy the item from you at $1.50 (and otherwise you will go to the next period). If you instead click on Exit the Market, you would get the value of the item ($1.00) for sure.

Each price between $1.50 and $0.00 is equally likely to be the price at which the computer decided to buy the item from you. In the first period, the price begins at $1.50. It decreases by $0.05 after each period that the computer does not buy the item. The task ends when the price decreases to $0.00 or you choose Exit the Market, whichever comes first.

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